8 Risk, Issue and Quality Management

8.1 Risk analysis and responses

ID / RISK	L×I	EMV (AUD)	OWNER / MITIGATION
R1 Incorrect answers	4×4=16	0.35 × $3,000 = $1,050	Ritish: Ground in approved FAQs; test; handoff on low confidence.
R2 Privacy disclosure	3×5=15	0.20 × $6,000 = $1,200	Deboprio: Minimise data; redact logs; restrict access; halt pilot if found.
R3 Prompt injection	3×5=15	0.25 × $4,000 = $1,000	Deboprio: Filter untrusted instructions; adversarial security tests.
R4 Content delays	4×3=12	0.35 × $2,000 = $700	Rohan: Content owner and freeze date; reduce initial FAQ set.
R5 Integration failure	3×4=12	0.25 × $3,000 = $750	Ganesh: Early API mock; rollback to widget-only pilot.
R6 Cost/latency spike	3×3=9	0.25 × $1,000 = $250	Ganesh: Usage caps; caching; monitor cost and response time.

The total EMV is AUD 4,950, and this falls within the suggested AUD 5,000 contingency. EMV is not the worst-case scenario; one bad incident may cost more than the contingency, and so there is a need to re-evaluate the budget before the pilot stage. In case of unsafe action and personal data leakages, the contingency will entail disabling the chatbot, keeping evidence of the incident, reporting to the sponsor, and provision of human assistance.

8.2 Issues and quality assurance

The owner of the identified issue logs it into the Jira tool with details of its severity, impact, the assignee, and a deadline for its resolution. Where there are privacy or security risks involved, they are escalated immediately to the sponsor and IT lead; functionality issues undergo triaging on a daily basis. Closing of the issue needs documented evidence of rectification followed by testing.

The quality gates include the following: (Q1) sponsor's approval of FAQ and rules for escalating issues; (Q2) clear requirements-traceability to relevant test cases; (Q3) accuracy assessment consisting of 100 questions; (Q4) adversarial testing, privacy testing, accessibility testing, and performance testing; and (Q5) pilot sign-off of the results with respect to O1-O4 objectives. Failure cases are logged by the test lead; release is prevented where there are any outstanding critical findings [3]–[6].